

Advanced Apex Project-1 (M.Sc. DS S AI)

Week-2 Project Outline

1. Project Title

Customer Behaviour Profiling & Micro-Segmentation

2. Problem Statement

Retail businesses generate large amounts of transactional customer data, but identifying meaningful customer groups and understanding shopping behavior remains a challenge. This project aims to analyze customer purchasing patterns and perform micro-segmentation using clustering techniques based on RFM metrics, Average Basket Size, and Product Diversity to identify distinct customer behavior profiles.

3. Business Goal / Objective

- Identify customer segments based on purchasing behavior.
- Understand customer spending and engagement patterns.
- Profile high-value, loyal, and churn-risk customers.
- Support targeted marketing and personalized offers.
- Improve customer retention and business decision-making

4. Dataset Details

- Dataset Name: UCI Online Retail II Dataset
- Source: Kaggle
- Dataset Link: <https://www.kaggle.com/datasets/jillwang87/online-retail-ii>
- Approximate Size: ~525,000 transaction records with 8 features
- Target Variable: Not applicable (Unsupervised Learning / Clustering Problem)

Important Features:

- InvoiceNo: Unique invoice or transaction number
- StockCode: Unique product/item code
- Description: Product description
- Quantity: Number of items purchased
- InvoiceDate: Date and time of transaction
- UnitPrice: Price per product unit
- CustomerID: Unique customer identifier
- Country: Customer country/location

5. Initial Understanding of the Dataset

- The dataset contains 525,461 retail transaction records with 8 features related to invoices, products, customers, pricing, and geographic information.
- The dataset includes both numerical and categorical variables:
 - Numerical: Quantity, UnitPrice
 - Categorical: InvoiceNo, StockCode, Description, CustomerID, Country
- InvoiceDate is currently stored as an object datatype and needs to be converted into datetime format for time-based analysis such as Recency calculation and purchase trend analysis.
- CustomerID contains 107,927 missing values, which is important because customer-level behavioral analysis and segmentation depend on customer identification.
- Description contains 2,928 missing values, indicating that some products do not have descriptions.
- Other columns do not contain missing values, showing relatively stable transactional information.
- The dataset contains:
 - 28,816 unique invoices
 - 4,632 unique products
 - 4,383 unique customers
 - Customers from 40 different countries
- The large number of unique invoices and products indicates rich transactional diversity suitable for customer behavior analysis and micro-segmentation.
- Multiple transactions by the same customers suggest the dataset is suitable for RFM analysis and customer profiling.
- The dataset represents real-world retail transactional data and may include cancellations, returns, duplicate records, and noisy data that require preprocessing before clustering and behavioral analysis.
- The presence of transactional, customer, and product-level information makes the dataset suitable for:
 - Customer segmentation
 - Behavioral profiling
 - RFM analysis
 - Spending pattern analysis
 - Customer retention analysis
 - Personalized marketing insights

6. Proposed Workflow

Data Loading → Data Understanding & Preprocessing → Exploratory Data Analysis (EDA) → Feature Engineering (RFM, Average Basket Size, Product Diversity) → Customer Behaviour Profiling → Customer Micro-Segmentation using Clustering → Cluster Evaluation & Interpretation → Business Insights & Visualization

7. Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

8. References / Dataset Sources

1. Kaggle – UCI Online Retail II Dataset

<https://www.kaggle.com/datasets/jillwang87/online-retail-ii>